

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	13 July 2026
Team ID	[To be filled by student/team]
Project Name	Voice-Based Concept Understanding Analyzer (VBCUA)
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template

Brainstorming provides a free and open environment that encourages every team member to participate in the creative thinking process that leads to problem solving. Volume was prioritised over value during the session: out-of-the-box ideas were welcomed and built upon, and all participants were encouraged to collaborate in developing a rich set of possible solutions for the Voice-Based Concept Understanding Analyzer (VBCUA).

This document records the outcome of the team's brainstorming session, the ideas generated, and the prioritisation exercise used to decide which ideas were carried forward into the implemented solution and which were deferred as future scope.

Reference: <https://www.mural.co/templates/brainstorm-and-idea-prioritization>

Step 1: Team Gathering, Collaboration and Select the Problem Statement

The team reviewed the customer problem statements defined during the Empathy and Define stages and consolidated them into a single guiding "How Might We" statement to focus the brainstorming session.

PROBLEM <i>How might we help students, self-learners, and evaluators objectively and efficiently assess spoken conceptual understanding, without relying entirely on manual review?</i>

Step 2: Brainstorm, Idea Listing and Grouping

During the brainstorming session, ideas were listed individually and then clustered into related groups. The table below presents the consolidated and grouped ideas generated for VBCUA.

Idea Cluster	Brainstormed Ideas
Speech Processing	- Convert spoken responses to text using OpenAI Whisper. - Allow real-time microphone recording directly in the browser. - Support multiple audio formats (WAV, MP3, M4A). - Add multi-language speech recognition support. - Apply audio pre-processing (noise reduction) using Librosa.
Semantic Evaluation	- Compare the transcript with a reference answer using Sentence Transformer embeddings and cosine similarity. - Support multiple acceptable reference answers per concept. - Highlight specific keywords or sub-concepts missing from the response.

	• Detect similarity between different students' responses.
Scoring & Feedback	• Convert similarity score into a percentage-based concept understanding score. • Assign a qualitative grade (Excellent / Good / Fair / Needs Improvement). • Generate descriptive feedback text for each grade tier. • Provide targeted, concept-specific improvement suggestions.
Reporting & Documentation	• Generate a downloadable PDF assessment report using ReportLab. • Include transcript, score, grade, feedback, and timestamp in the report. • Maintain a score history or performance dashboard across attempts. • Export results in additional formats such as CSV.
Platform & Usability	• Build an interactive web interface using Streamlit. • Provide a sidebar with project objectives, formats, and technology stack. • Add drag-and-drop audio upload with format validation. • Introduce user authentication and role-based access (student / evaluator). • Deploy the application to the cloud for remote access.

Step 3: Idea Prioritization

The grouped ideas were plotted against two criteria: Importance (value delivered to the student/evaluator) and Feasibility (achievability within the project's time, skill, and resource constraints). Ideas rated high on both criteria were selected for implementation in the current version of VBCUA; the remaining ideas were retained as future scope or deprioritised.

↑ Importance → High Feasibility	↑ Importance → Low Feasibility
High Importance / High Feasibility — Do Now (Implemented) • Speech-to-text transcription using OpenAI Whisper • Semantic similarity comparison using Sentence Transformers • Automated concept understanding score and grade • Downloadable PDF assessment report • Streamlit web interface for upload and results	High Importance / Low Feasibility — Consider (Future Scope) • Real-time microphone recording • Support for multiple reference answers per concept • Multi-language speech recognition • Keyword-level feedback on missing concepts • Student performance dashboard / score history
Low Importance / High Feasibility — Quick Wins • Drag-and-drop upload styling refinements • Sidebar project information panel • Configurable UI theme colours	Low Importance / Low Feasibility — Deprioritise • Cross-student plagiarism / similarity detection • Native mobile application • User authentication and role management • Cloud deployment and database integration